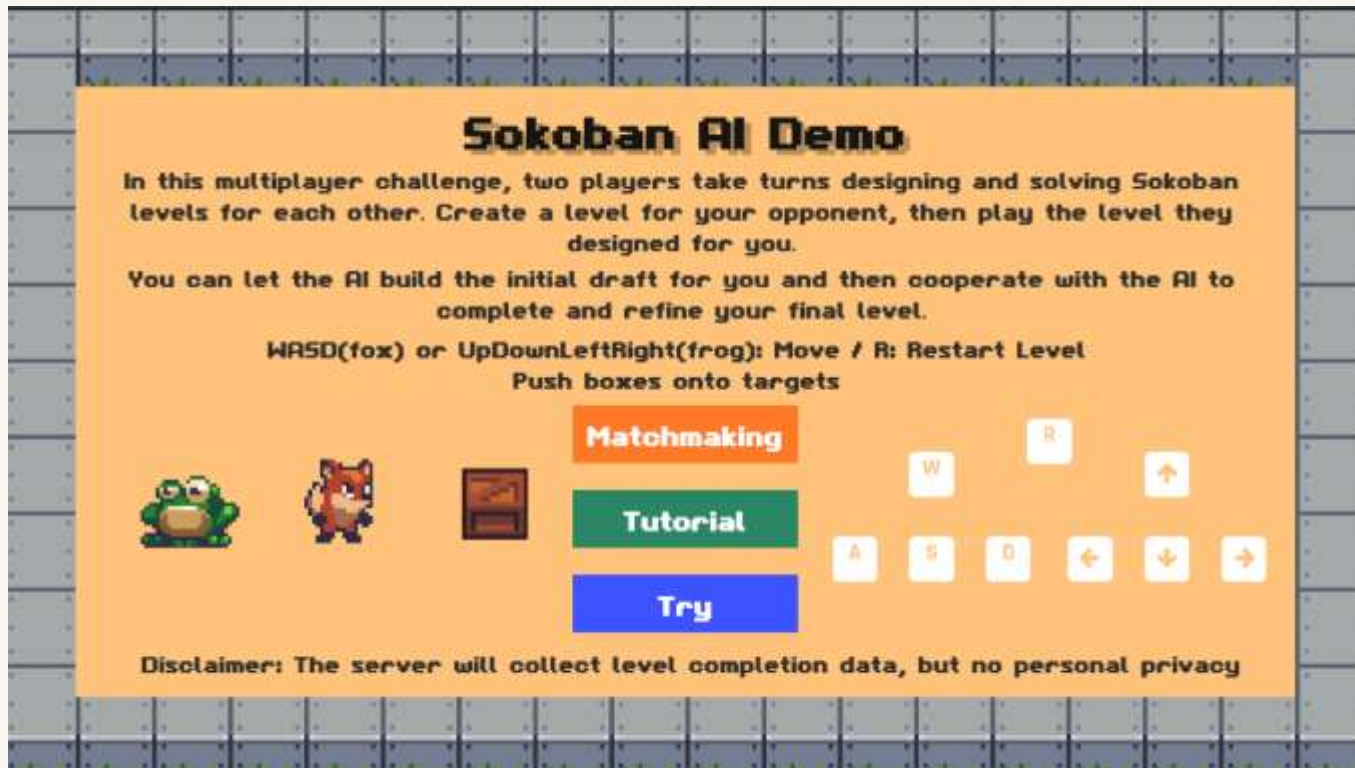


Sokoban AI Demo 教程 / Tutorial

多人共创推箱子挑战 | Bilingual Guide for Multiplayer Co-Creation



你将做什么？ / What will you do?

- 与另一位玩家互相设计并游玩 Sokoban 关卡。
/Design and play Sokoban levels with another player.
- 先创建或加入房间，再进入共创地图阶段。
/Create or join a room, then enter co-creation.
- 和 AI 一起修改、试玩、验证并确认最终关卡
/Revise, test, verify, and confirm the final level with AI.
- 最后游玩对手的关卡，并完成结果记录。
/Finally, play your opponent level and record results.



关键词 / Keywords:

狐狸 = 试玩自己的关卡 / Fox = test your own level

青蛙 = 游玩对手关卡 / Frog = opponent level

箱子 = 推到目标点 / Box = push to the target

1. 基础规则与操作 / Basic Rules & Controls

先认识角色、按键和时间限制 / Learn the characters, controls, and time limits first

角色与操作 / Roles & Controls

- 狐狸：试玩自己设计的关卡。
/ Fox: test your own level.
- 操作：W / A / S / D 移动，R 重新开始。
/ Move with W/A/S/D; press R to restart.
- 青蛙：游玩对手设计的关卡。
/ Frog: play your opponent level.
- 操作：方向键 ↑ ↓ ← → 移动，R 重新开始。
/ Move with arrow keys; press R to restart.
- 核心目标：把箱子推到目标点。
/ Goal: push boxes onto targets.



狐狸 / Fox
WASD



青蛙 / Frog
方向键 / Arrow keys



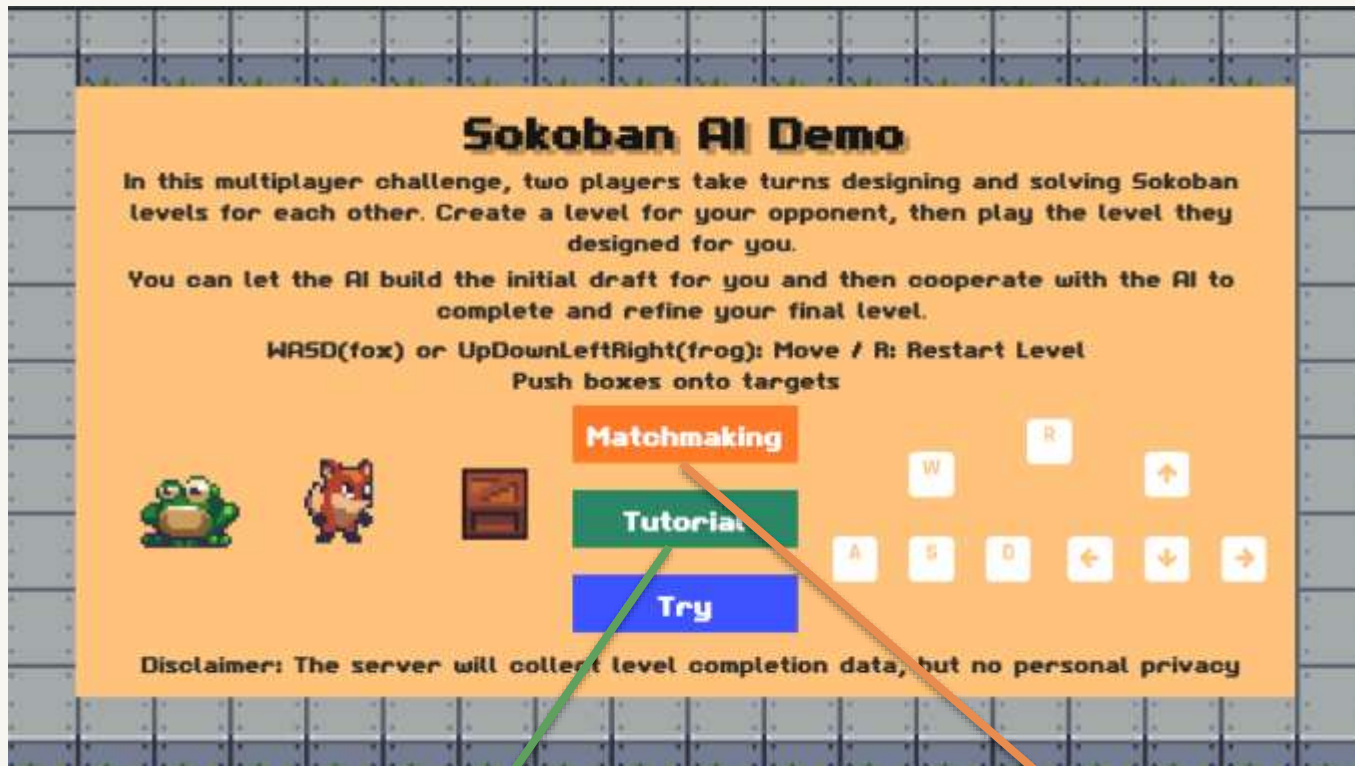
箱子 / Box
推到目标点 / Push to target

共创阶段限时 10 分钟
Co-creation limit: 10 min

游玩对手关卡限时 30 秒
Opponent level limit: 30 sec

2. 从主菜单开始 / Start from the Main Menu

阅读说明后，进入匹配流程 / After reading the instructions, enter matchmaking



教程按钮
Tutorial button

匹配入口
Matchmaking entry

主菜单说明 / Main Menu Guide

- Tutorial: 查看本教程。
/ View this tutorial.
- Server: 连接服务器。
/ Connect to the server.
- Matchmaking: 进入匹配大厅。
/ Enter the matching lobby.
- Try: 可进入本地试玩。
/ Try local practice.
- 底部说明: 系统会收集关卡完成数据, 但不采集个人隐私。
/ Footer note: completion data is collected, but personal privacy is not.

3. 匹配大厅：先创建房间码，再加入房间 / Matchmaking: Create a Code, Then Join

一名玩家创建房间，另一名玩家使用房间码加入 / One player creates a room; the other joins with the code

01 CREATE

Create a room

Open a private match and share the generated code with your opponent.

ROOM CODE APPEARS HERE

CREATE ROOM

步骤 1：点击 CREATE ROOM
Step 1: Click CREATE ROOM

生成房间码并分享给对手
/ Share the room code with your opponent

02 JOIN

Join a room

Enter the six-digit room code shared by the room host.

JOIN ROOM

步骤 2：输入 6 位房间码
Step 2: Enter the 6-digit code

点击 JOIN ROOM 进入房间
/ Click JOIN ROOM to enter

注意：你们两位玩家需要进入同一个房间，随后才能开始后续共创与交换关卡。 / Note: both players must enter the same room before co-creation and level exchange can begin.

4. 创建首版 Draft / Create the Starting Draft

回答四个中立问题，帮助 AI 理解你的初始设计方向 / Answer four neutral questions to help AI understand your initial design direction

Draft

Answer four quick questions. The AI will summarize your choices and give you a suggestion. You can also adjust the parameter by yourself.

DESIGN BRIEF

How much inspection should the map require before the first move?

- Little inspection before acting
- Some inspection of boxes, goals, and passages
- Broader inspection and planning before acting
- No preference

AI reflection will appear here.

Layout: Balanced (left arrow) (right arrow)

Difficulty: Medium (left arrow) (right arrow)

AI difficulty suggestion will appear here.

AI layout suggestion will appear here.

CONFIRM

调节参数
Adjust parameters

选择答案
Select answers

Draft 第一步 / Draft Step 1

- 阅读每个问题，选择最符合你想让对手获得的体验的选项。
/ Choose the option that best matches the experience you want your opponent to have.
- 这些回答只用于帮助 AI 生成初始建议，不是最终限制。
/ These answers guide the AI's first suggestion; they are not final limits.
- 确认前，你仍可以手动调整 Layout 与 Difficulty。
/ Before confirming, you can still adjust Layout and Difficulty manually.
- 没有明确想法时，可以选择 No preference。
/ If you have no clear preference, choose No preference.

4. 创建首版 Draft / Create the Starting Draft

查看 AI 总结和参数建议，必要时手动调整 / Review the AI summary and parameter suggestions, then adjust if needed

Draft

Answer four quick questions. The AI will summarize your choices and give you a suggestion. You can also adjust the parameter by yourself.

DESIGN BRIEF

AI is reviewing your initial-map preferences.

- Short routes connecting nearby decisions
- Mostly direct routes with some detours
- Longer routes with exploration or return paths
- No preference

AI reflection: Your choices suggest you want to start moving quickly instead of inspecting everything first. It sounds like some pushes should depend on position or order. My read is that important spots are spread across a few connected areas with longer routes between decisions.

Layout ◀ Open ▶ AI difficulty suggestion: Easy
Difficulty ◀ Easy ▶ AI layout suggestion: Open

CONFIRM

AI 总结与推荐
AI summary & suggestion

确认并继续
Confirm & continue

AI 推荐说明 / AI Recommendation Guide

- AI 会根据你的回答生成 reflection，总结它理解到的设计方向。
/ The AI generates a reflection summarizing its read of your design direction.
- 查看 AI 推荐的 Difficulty 和 Layout 是否符合你的想法。
/ Check whether the suggested Difficulty and Layout match your intention.
- 你可以接受建议，也可以手动调整参数。
/ You may accept the suggestion or adjust the parameters manually.
- 只有当首版方向合适时，再点击 CONFIRM。
/ Click CONFIRM only when the starting direction feels appropriate.
- 这只是首版草案，之后仍会进入 AI 共创修改阶段。
/ This is only the starting draft; you will still revise it with AI later.

5. 打开 AI 共创 Lab / Open the AI Co-Creation Lab

首版 Draft 已同步，打开浏览器进入持续共创 / The starting Draft is synchronized; open the browser workspace to continue co-creation



打开共创 Lab
Open co-creation lab

进入共创说明 / Entering Co-Creation

- 当前首版 Draft 已同步到独立的 AI 共创网页。
/ The current starting Draft is synchronized to the separate AI co-creation webpage.
- 点击 OPEN CO-CREATION LAB，在浏览器中打开共创空间。
/ Click OPEN CO-CREATION LAB to open the workspace in your browser.
- 在网页中查看当前 Stage，并通过多轮对话与 AI 讨论和修改。
/ Review the current Stage and discuss and revise it with AI over multiple turns.
- 提交最终设计意图后，只有最终确认的 Stage 会同步回 Unity。
/ After submitting your final design intention, only the final confirmed Stage is synchronized back to Unity.

6. 检查 AI 生成的首版 Draft / Review the AI-Generated Starting Draft

检查地图、选择会话语言，并决定重新生成或进入共创 / Check the map, choose the session language, then regenerate or enter co-creation



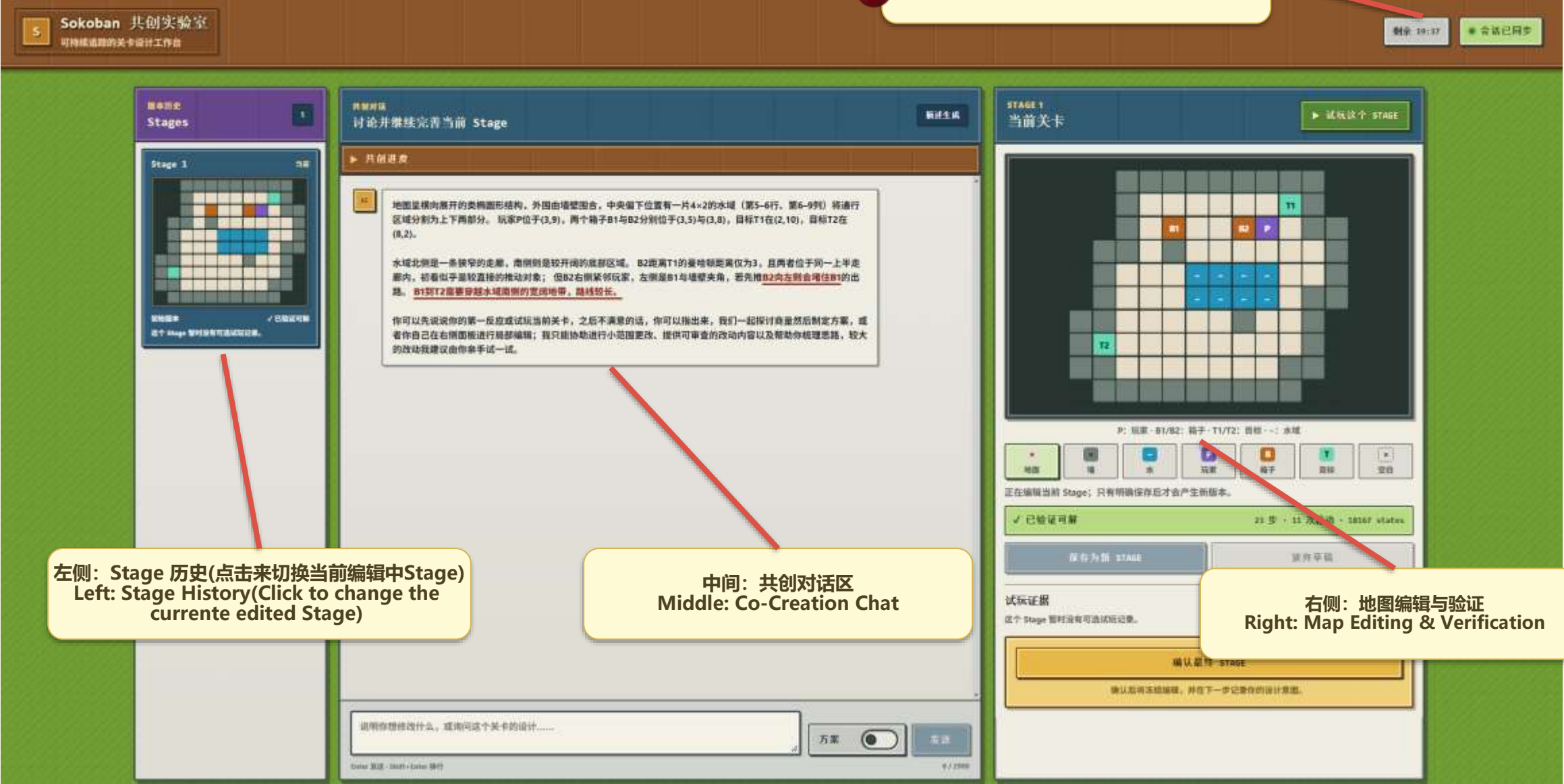
首版 Draft 操作说明 / Starting Draft Guide

- Unity 已根据 AI 规划生成并验证首版 Draft；进入共创后，当前 Draft 会成为 Stage 1。
/ Unity has generated and validated the starting Draft from the AI plan; it becomes Stage 1 when co-creation begins.
- 先检查地图预览和可解性结果。
/ Review the map preview and solvability result first.
- 不满意时点击“重新生成”；满意后选择本次共创会话语言。
/ Click Regenerate if needed; when satisfied, choose the language for this co-creation session.
- 进入共创后语言不可再切换，并开始共创倒计时。
/ Once co-creation starts, the language cannot be changed and the co-creation timer begins.
- 点击“进入共创流程”开始 Stage 1。
/ Click Enter Co-Creation to begin Stage 1.

7. 共创阶段总览 / Co-Creation Stage Overview

与 AI 讨论并持续完善当前 Stage / Discuss with AI and refine the current Stage

注意有限时
Attention for the time limit



8. 共创界面怎么用 / How to Use the Co-Creation Interface

看懂三个核心区域 / Understand the three core areas



Stage 历史 / Stage History

- 查看历史版本。
/ View historic versions.
- 覆盖当前 Stage。
/ Cover the current Stage.



共创对话区 / Co-Creation Chat

- 展开“共创进度”，查看 AI 对你设计意图的推测。
/ Co-Creation Progress shows the AI's inferred design intent.
- “方案”按钮：开启后进入方案流程，回答AI的问题并等待AI给出修改方案。
/ “Plan” button: Once enabled, it starts the planning process. Answer the AI's questions and wait for it to provide a revision plan.



地图编辑区 / Map Editing Area

- 查看并编辑当前关卡。
/ View and edit the current level.
- 放置地面、墙、水、玩家、箱子、目标等元素。
/ Place floor, wall, water, player, box, and target tiles.

8. 共创界面怎么用 / How to Use the Co-Creation Interface

看懂三个核心区域 / Understand the three core areas



AI 对话会出现五类卡片 (第五种为报错)
Five AI chat card types may appear; the fifth is error

共创对话区 / Co-Creation Chat

- 描述想法、表达不满意之处。
/ Describe ideas and concerns.
- 有下划线的文字描述可以点击查看右侧地图的路线引导箭头。
/ Underlined text can be clicked to display route-guidance arrows on the map to the right.
- AI 修改能力有限, 建议手动修改。(AI能够帮助你修改:水域,内部墙体)
/ Manual edits are recommended.



出现于玩家与 LLM 相互质疑流程, 用于讨论和澄清意图
Used during player-LLM challenges to discuss and clarify intent



AI 揣测你的修改意图
AI guesses your revision intent



AI 帮助你进行更改
AI helps make changes

自己实行更改 (推荐)
Make changes yourself (recommended)



看懂三个核心区域 / Understand the three core areas

看懂三个核心区域 / Understand the three core areas



- 箱子,起点和终点数量固定为2
/ The number of boxes, starting points, and goals is fixed at 2.
- 如果需要改动箱子,外壳,起点和终点的位置,请手动更改并让AI检验是否可解
/ If you need to change the positions of the boxes, shell, starting points, or goals, please adjust them manually and let the AI verify whether the level is solvable.

可选择的覆盖瓦片 Selectable replacement tiles

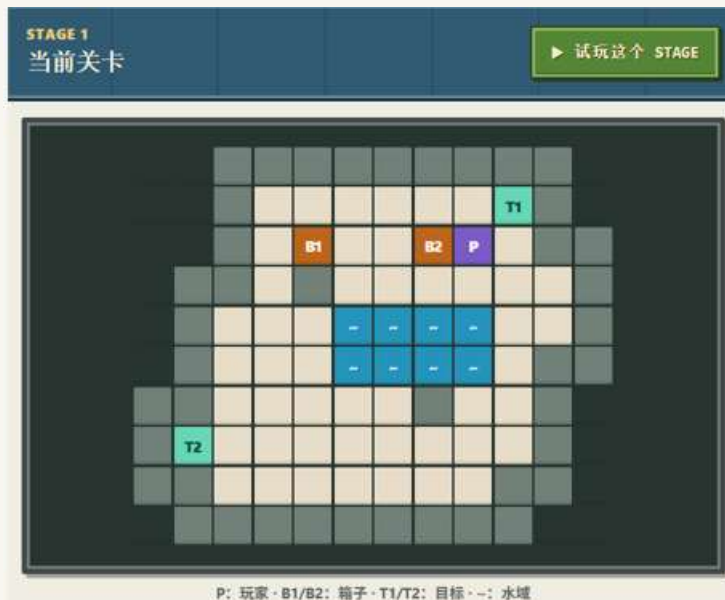
先选择要覆盖的瓦片作为画笔，再在图中点击替换瓦片
Choose a replacement tile as the brush, then click tiles on the map to replace them

修改后点击保存为新的 STAGE，再进入可解性检验
After editing, save as a new STAGE and run the solvability check

你可以在任意STAGE确认为最终STAGE
You can confirm as the final stage at any historical stage

9. 试玩、验证与确认 / Playtest, Verify & Confirm

在确认前，请尽量自己体验和检查关卡 / Before confirming, try and check the level yourself



试玩当前 Stage / Playtest Current Stage

- 点击“试玩这个 STAGE”。
/ Click “Try this STAGE”.
- 试玩自己设计的关卡时，你操控狐狸。
/ When testing your own level, you control the fox.

验证与保存 / Verify & Save

- 系统会显示关卡是否可解。
/ The system shows whether the level is solvable.
- 可查看步数、推箱次数等信息。
/ Check steps and box-push counts.
- 满意后可保存为新 Stage。
/ Save as a new Stage when satisfied.

确认最终关卡 / Confirm Final Level

- 点击“确认最终 STAGE”。
/ Click “Confirm Final STAGE”.
- 确认后冻结编辑，并给对手发一条信息。
/ Editing freezes, then send a message to your opponent

10. 游玩对手的关卡 / Play Opponent Level

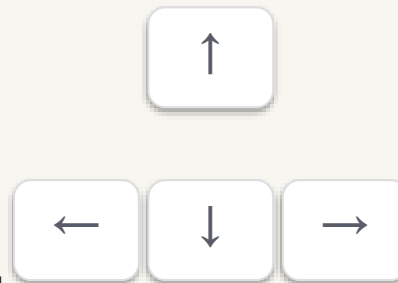
双方提交后，你将进入对手设计的关卡 / After both players submit, you enter your opponent level

挑战规则 / Challenge Rules

- 你将游玩另一位玩家设计的关卡。
/ You will play another player level.
- 此时可操控角色是青蛙。
/ You control the frog.
- 使用方向键 ↑ ↓ ← → 移动。
/ Move with the arrow keys.
- 本阶段限时 30 秒。
/ This stage is limited to 30 seconds.
- 目标仍然是把箱子推到目标点。
/ The goal is still to push boxes onto targets.



青蛙 = 对手关卡
Frog = opponent level



重要 / Important:
30 秒内尽力完成对手关卡
Try your best to finish within 30 seconds

8. 完整流程回顾 / Full Process Recap

开始前请再看一遍整体顺序 / Review the overall sequence before starting

1. 点击 Matchmaking并创建首版draft / Click Matchmaking and create draft

2. 创建房间码，另一位输入加入 / Create a room code; the other player joins

3. 与 AI 讨论并修改 Stage（20 分钟） / Discuss with AI and revise the Stage (20 min)

4. 试玩自己的关卡（狐狸 / WASD） / Test your own level (Fox / WASD)

5. 查看验证，保存新 Stage，并确认最终 STAGE / Verify, save, and confirm the final STAGE

6. 交换关卡并游玩对手关卡（青蛙 / 方向键 / 30 秒） / Swap and play opponent level (Frog / Arrow keys / 30 sec)

7. 完成记录与结果页面 / Complete the record and results page

祝你和队友顺利完成共创，并在挑战中获得有趣的体验！
Good luck co-creating with your teammate, and enjoy the challenge!